PLEASE NOTE: THIS IS A PROVISIONAL SYLLABUS

## COURSE INFORMATION

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| **COURSE TITLE** | **Applications of Artificial Intelligence Models** |
| **COURSE NUMBER** | **SCH-MGMT 661(In-person section[1])** |
| **CLASS MEETINGS** | When: **Tu/Th 10:00 – 11:15 a.m.**<br>Where: **RM G22** |
| **INSTRUCTORS** | Indika Dissanayake, Ph.D. (RM 335) idissanayake@isenberg.umass.edu |
| **OFFICE HOURS (TENTATIVE)** | When: **Tu 11:30 a.m. – 12:30 p.m., Th 5:00 – 6:00 p.m., or by appointment**<br>Where: **Zoom** |
| **COURSE DESCRIPTION** | The course prepares students to consider the application of AI in business. Students will learn to use standard machine learning (ML) models in Python and understand how to apply them across a range of business functions. Students will propose and implement an ML project. |

## STUDENT LEARNING OBJECTIVES

Upon successful completion of this course, students will be able to:

- Gain a practical understanding of core concepts, models, and applications of Artificial Intelligence (AI) and Machine Learning (ML), including key areas such as neural networks, deep learning, natural language processing (NLP), computer vision, and generative AI.
- Recognize the importance of data acquisition and preprocessing in building effective AI/ML models.
- Define business problems and apply appropriate ML/AI approaches, selecting tools and models to explore and develop potential solutions.
- Communicate the business implications of AI outcomes, including both opportunities and limitations.
- Build experience working with real-world data through hands-on modeling and analysis, by completing structured exercises and a group or individual project.

## HOW IS THIS COURSE STRUCTURED?

The course includes a mix of lectures, discussions, problem solving exercises and other experiential learning activities such as exploring content-related concepts through hands-on exercises. These learning activities will provide you with practical knowledge of how to apply AI/ML methods and techniques to solve a variety of business operations problems.

## COURSE MATERIALS, TOOLS, & OTHER RESOURCES

**Required textbook**: Deep Learning with Python (**DLWP**). 3rd Edition by Francois Chollet and Matthew Watson. Manning Publications., 2025

**Other course materials:** All other course materials will be provided online via canvas.

[1] Note: If a policy or requirement applies specifically to either the in-person or online section, it will be clearly indicated. Otherwise, the information is relevant to both sections.

**Software Tools:** The course will utilize Google Colab with Jupyter notebooks.
**Additional resources (optional):**

- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. 3rd edition by Aurélien Géron, O'Reilly Media, Inc., 2023
- Generative Deep Learning. 2nd edition by David Foster, O'Reilly Media, Inc., 2023
- Google ML Crash course
- DataCamp
- DeepLearning.AI
- Hugging Face – AI community.

## EVALUATION & GRADING SCALE

The course has a variety of activities to allow students to become familiar with Artificial Intelligence (AI) and Machine Learning (ML) technologies and how they augment business operations. Upon completio of this course, students will be awarded a letter grade based on the following criteria.

| Activity | Points |
|---|---|
| Weekly Exercises & Quizzes | 20% |
| Assignments | 30% |
| Preparation, Participation, and Engagement | 25% |
| Team/ Solo Project | 25% |

Grading Scale:

| Grade | Points |
|---|---|
| A | 93 to 100 |
| A- | 90 to < 93 |
| B+ | 87 to < 90 |
| B | 83 to <87 |
| B- | 80 to <83 |
| C+ | 77 to <80 |
| C | 73 to <77 |
| F | <73 |

## TYPE OF ASSESSMENT & LATE WORK POLICY

**PREPARATION**

Class preparation includes assigned readings, reviewing course modules available on Canvas, and materials from other learning platforms such as DataCamp. Students are expected to review these resources, including lecture notes and recordings when available, to develop a conceptual understanding of the material before class.

**ASSIGNMENTS**

Assignments are an indispensable part of this course and are important for learning the tools and methods taught in this class. You will work on and submit several individual assignments.

- Submission of assignments will be done electronically via Canvas.
- Students are expected to submit their own work for individual assignments and to work with their team and contribute to their group work.
- Students are expected to submit professional work; all graphs, computer output, tables etc. must be organized and clearly labeled.

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| **QUIZZES** | Quizzes will be given throughout the semester outside of class time. The purpose of the quizzes is to encourage regular review of the material and to reinforce your understanding of the subject matter. You will submit quizzes online via our course Learning Management System, Canvas Learn. |
| **IN-CLASS/ ONLINE WEEKLY EXERCISES** | Students will complete several hands-on exercises during class sessions and submit them via the course Learning Management System, Canvas Learn, in the in-person section. In the online section, students are expected to review the weekly materials and complete the assigned weekly exercises on time. |
| **TEAM PROJECT** | Students will typically work in teams of 3-4 to complete a project focused on applying AI models to a business problem. The project should start early in the course and will include multiple deliverables, including a final presentation. Group projects cannot be completed individually. This is the default option for the in-person section. Students in the online section may choose to complete either a team project or an individual project. |
| **ENGAGEMENT IN THE COURSE** | Throughout the semester, you are expected to actively engage and participate in all aspects of the course. Your engagement grade will be based on attendance and participation in course activities. This includes but is not limited to: completing prep-work before class, contributing to discussions in class or on Canvas discussion boards, responding to questions through the Top Hat system, and providing peer feedback. Overall, we will consider whether you are simply meeting the minimum requirements (e.g., submitting assignments and quizzes) or making a meaningful effort to fully engage with the learning process.<br><br>Please don't be shy about speaking up. You won't be penalized for making mistakes. In fact, there may often be more than one valid answer. Your participation not only enhances your own learning but also contributes to a more dynamic and rewarding experience for the entire class. |
| **LATE POLICY** | All assessments in this course, including assignments, projects, quizzes, and related deliverables, are due on the date and time specified in Canvas. A late submission will be accepted for up to one week after the deadline, and a 2% deduction will be automatically applied for each day the work is late. One week after the deadline, the assessment will close in Canvas, and no further submissions will be accepted. In rare and exceptional circumstances, the late penalty may be waived, provided the student communicates with the instructor prior to the assessment deadline. |

## COMMUNICATION

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| **COMMUNICATION** | Students are expected to maintain an active UMass email account and to check both their email and course website (Canvas) daily. Generally, students can expect a response to emails within 48 hours on weekdays. However, if students wait until the last minute to ask assignment-related questions, it is possible that they may not receive a response before the deadline. Assignments will be posted at least a week in advance to provide students with adequate time to complete them and seek clarification if needed. |
| **Canvas** | To complement and enhance our class meetings Canvas will be used as our administrative assistant, communication facilitator and |

information center. Course materials (e.g. links to videos, slides), announcements, communication, submission of work, etc. will be done via our Canvas course.
To access our course on Canvas, go to: https://www.umass.edu/it/canvas. **Please check Canvas regularly**.

**ZOOM** You must use your UMass Zoom account for any online office hours or for Zoom meetings. In case of extreme weather, the instructor may choose to hold in-person class over Zoom.

## ARTIFICIAL INTELLIGENCE (AI) POLICY

**AI is encouraged with certain tasks and with attribution:**
You can choose to use AI tools to help brainstorm assignments or projects, to revise existing work you have written, or as a learning tool. When you submit your assignment, I expect you to clearly attribute what text or code was generated by the AI tool or include prompt. (e.g., AI-generated text appears in a different colored font, quoted directly in the text, or use an in-text parenthetical citation)

## ATTENDANCE POLICY

***In-person section:***
Attendance is not optional for all scheduled in-person class meetings. If you are enrolled in an in-person section, you are expected to attend every class. If you must miss class due to an emergency, you must notify the instructor as soon as it is reasonably possible to do so. For absences known in advance (e.g., university-sponsored events, religious observances), inform the instructor ahead of time, ideally at least one week in advance. Make-up work will be allowed only for excused absences with a valid reason and timely communication.

***All Sections (In-person and online):***
You are expected to complete all required deliverables on time. If an extension is granted due to a valid reason communicated before the deadline, the missed work must be submitted by the agreed-upon due date—no exceptions. Do not wait until the end of the semester to resolve missed work.

## ACADEMIC HONESTY STATEMENT

Since the integrity of the academic enterprise of any institution of higher education requires honesty in scholarship and research, academic honesty is required of all students at the University of Massachusetts Amherst. Academic dishonesty is prohibited in all programs of the University. Academic dishonesty includes but is not limited to: cheating, fabrication, plagiarism, and facilitating dishonesty. Appropriate sanctions may be imposed on any student who has committed an act of academic dishonesty. Instructors should take reasonable steps to address academic misconduct. Any person who has reason to believe that a student has committed academic dishonesty should bring such information to the attention of the appropriate course instructor as soon as possible. Instances of academic dishonesty not related to a specific course should be brought to the attention of the appropriate department Head or Chair. Since students are expected to be familiar with this policy and the commonly accepted standards of academic integrity, ignorance of such standards is not normally sufficient evidence of lack of intent.
(http://www.umass.edu/dean_students/codeofconduct/acadhonesty/).

## ACCOMMODATION STATEMENT

The University of Massachusetts Amherst is committed to providing an equal educational opportunity for all students. If you have a documented physical, psychological, or learning disability on file with Disability Services (DS), you may be eligible for reasonable academic accommodations to help you succeed in this course. If you have a documented disability that requires an accommodation, please notify me within the first two weeks of the semester so that we may make appropriate arrangements. For further information, please visit Disability Services. (https://www.umass.edu/disability/)

## TITLE IX STATEMENT

In accordance with Title IX of the Education Amendments of 1972 that prohibits gender-based discrimination in educational settings that receive federal funds, the University of Massachusetts Amherst is committed to providing a safe learning environment for all students, free from all forms of discrimination, including sexual assault, sexual harassment, domestic violence, dating violence, stalking, and retaliation. This includes interactions in person or online through digital platforms and social media. Title IX also protects against discrimination on the basis of pregnancy, childbirth, false pregnancy, miscarriage, abortion, or related conditions, including recovery. There are resources here on campus to support you. A summary of the available Title IX resources (confidential and non-confidential) can be found at the following link: **https://www.umass.edu/titleix/resources**. You do not need to make a formal report to access them. If you need immediate support, you are not alone. Free and confidential support is available 24 hours a day / 7 days a week / 365 days a year at the SASA Hotline 413-545-0800

## SYLLABUS CHANGES

All syllabus (or any other) changes will be announced in the online course space, during class time and/or via E-mail. It is the student's responsibility to ensure that they are up to date.

Tentative Schedule

| Module | Topic | Class Prep (Tools, Chapters, and Other Resources) |
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| 1 | **Introduction** (AI, ML, and Deep Learning Overview) | DLWP – Ch 1 |
| 2 | **Data Representation and Exploratory Data Analysis** (EDA Review & Intro. to Tensor Flow)<br><br>*Case/ Application 1: EDA with Airbnb Listing Data (Inside Airbnb)* | Refresh skills/ tools: Data pre-processing and visualization with Pandas, seaborn, matplotlib<br><br>DLWP Ch. 2.2: Data representation (Tensor Flow framework)<br><br>DLWP Ch 6: Workflow of ML |
| 3 | **Regression with Keras + Intro to Neural Networks (NN)**<br><br>*Case/Application 2: Price Prediction Model with Airbnb Listing data using Keras Sequential model* | DLWP Ch. 3.6 (introduction to Keras), 4.3 (Regression with Keras Sequential)<br><br>Optional:<br>Linear regression (Google Crash Course) |
| 4 | **Classification with Keras (Activation Functions, Loss Functions, Optimizers, and Backpropagation)**<br><br>*Case/Application 3a: Movie Reviews Binary Classification Model with IMDb Dataset using Keras Sequential Model* | DLWP Ch. 4.1- 4.2<br><br>Optional:<br>Classification (Google Crash Course)<br>Neural Networks (Google Crash Course) |
| 5 | **Improving Model Performance**<br><br>*Case/Application 3b: Improve Movie Reviews Binary Classification Model using Regularization Techniques* | DLWP Ch. 5 |
| 6 | **Natural Language Processing Part I: Concepts and Pre-Processing**<br><br>Text representations (e.g., Bag-of-Words & TF–IDF), word embeddings, and preprocessing techniques<br><br>*Case/Application 4: Preprocessing Airbnb Reviews for Downstream Tasks* | DLWP Ch. 14 |
| 7 | **Natural Language Processing Part II: Topic Modeling**<br><br>Traditional and Modern Topic Modeling Techniques<br><br>*Case/Application 5: Identify Themes in Airbnb Guest Reviews* | DataCamp Course: Working with Hugging Face |

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| 8 | **Natural Language Processing Part III: Hugging Face Transformers pipeline for NLP**<br><br>Sentiment analysis, Named entity recognition (NER) Part-of-speech tagging (POS), Language Translation, Text Summarization, Zer-shot classification<br><br>*Case/Application 6: Application of HF pipeline to analyze Airbnb Guest Reviews* | Hugging Face Transformers Pipeline function |
| 9 | **NN Recap+ Intro to Convolutional Neural Networks (CNNs)**<br><br>*Case/Application 7: Image Classification with MNIST* dataset | DLWP Ch. 7<br><br>DataCamp Course: Introduction to Deep Learning with Keras |
| 10 | **CNNs + Transfer Learning (**week or two**)**<br><br>*Case/Application 8a: Image Classification with CIFAR-10 dataset*<br>*Case/Application 8b: Transfer Learning with MobileNet (CIFAR-10)* | DLWP Ch. 8 |
| 11 | **Object Detection**<br><br>*Case/Application 9: Safety Helmet Detection with YOLO* | DLWP Ch. 12 |
| 12 | **Image Generation & Captioning (Hugging Face + Gradio Interface)** | Hugging Face Transformers ("text-to-image", image-to-text")<br><br>Gradio App Tutorials |
| 13 | **Intro to LangChain** (Tentative – Based on Course Progress) | |
| | **Projects and Presentations** | |

## Dataset Summary:

| Modality | Dataset(s) | Topics |
|---|---|---|
| **Tabular** | Airbnb listings (Inside Airbnb) | Regression, classification, project foundation |
| **Text** | Airbnb guest reviews, IMDb dataset | NNs, Binary Classification, Sentiment, topic modeling, NER, zero-shot classification |
| **Vision** | MNIST, CIFAR-10, Safety Helmet Images | CNNs, transfer learning, object detection |
| **Multimodal** | Student-generated apps using Airbnb + Gradio & LangChain (tentative) | Deployment and interaction across data types |

***Note: Assessment deadlines will be posted on Canvas.***